Supply Chain Finance Innovation Using Blockchain

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***Abstract*—Blockchain, the underlying technology of digital currency, such as Bitcoin, has the characteristics of decentralization, stability, security, anonymity, and nontampering. Blockchain is being applied in an increasing number of fields. Supply chain finance is a financing model in which banks connect core enterprises with upstream and downstream enterprises to provide flexible financial products and services. In the traditional supply chain finance, the core enterprises with strong competitiveness and a large scale play an irreplaceable role in managing the supply chain information flow, logistics, and capital flow, which results in inequality and information asymmetry. Fraud in supply chain finance is also very serious. To solve the problems of the traditional supply chain finance, in this article, we build a new type of supply chain financial platform that uses blockchain technology to manage the whole process. This supply chain financial platform solves the problem of nontrust among the participants in the supply chain, improves the efficiency of the capital flow and information flow, reduces costs, and provides better financial services to the relevant parties in the supply chain. To protect users' privacy, we propose a new method of using homomorphic encryption in the blockchain to meet the needs of sensitive data privacy protection in supply chain financial scenarios.**

***Index Terms*—Consortium blockchain, fintech, supply chain financing.**

I. Introduction

SUPPLY chain finance is originated from trade finance. In the 1940s, Saulnier studied the development of the accounts receivable financing business in American trade finance. However, at this time, accounts receivable financing was financed mainly through the bill discount business, and there were no obvious characteristics of supply chain finance [1]. Then, Koch, Sutkowski, and Wessman discussed logistics finance as represented by inventory financing [2]–[4]. At the end of the 20th century, Hartley *et al.* proposed the concept of supply chain financing and introduced the financing design of prepaid accounts. Randall and Farris defined the supply chain finance as the cooperation of upstream and downstream enterprises in the production of the supply chain in terms of capital flow through which the overall cost of supply chain production can be reduced and supply chain revenue can be enhanced [5]. At present, supply chain finance plays an important role in serving the real economy and helping small to medium-sized enterprises (SMEs) to finance. With the continuous innovation of the business model of supply chain finance and the continuous advancement of information construction, researchers are paying increasing attention to the role of information technology in supply chain finance. Supply chain finance has entered the stage of informatization and intellectualization [6]. The problem of information asymmetry among the relevant parties in the supply chain is alleviated and the cost of information circulation is reduced via the construction of the enterprise information system. Therefore, an Internet-based supply chain financial system can provide a convenient and rapid means for financing or offering other financial services of supply chain enterprises [7].

As the world's leading producer and consumer of commodities, China plays a pivotal role in the global commodity sector. However, as the global economy enters a downward adjustment cycle, defaults and fraudulent loans are becoming frequent, and the credit crisis of commodities has gradually emerged. Risk events, such as repeated pledges and empty pledges caused by asymmetric information, have seriously hampered the mutual trust of all parties involved in financing, which makes it difficult to ensure management integrity and warehousing supervision. The financial order has been disrupted, bad debts on bank loans have increased, and banks have tended to reduce loans and accelerate the collection of loans. At the same time, the circulation of bulk goods is difficult, traders have tight capital chains, loans are more difficult to obtain, and business development is in trouble. Blockchain technology can effectively solve the information asymmetry problem in warehousing, logistics, and supervision.

As the underlying technical framework of digital currency, blockchain technology has the characteristics of decentralization and nontamperability [8]–[10]. The nodes in a blockchain system reach a consensus based on not being able to fully trust the remaining nodes and jointly completing the distributed storage and confirmation of transactions. The confirmation and record of transactions in a traditional trading system are completely dependent on a trusted organization. This centralized system faces many problems in terms of transaction costs, efficiency, and security. Blockchain technology has changed this situation. The nodes in the blockchain have equal status, and consensus can be reached without a third organization. The blockchain technology system is no longer limited to digital currency and has begun to be applied to the Internet of things (IoT) [11], [12], transportation services [13], energy networks [14], [15], medical data [16], the banking industry [17], and other fields. In the supply chain area, many researchers have found that the

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blockchain and the supply chain are much matched. Toyoda *et al.* [18] used Bitcoin and radio frequency identification to build a novel product ownership management system for anti-counterfeits in the post supply chain. Liu *et al.* [19] proposed a blockchain-based tobacco supply chain financial service platform framework and provided a simple analysis of a business model. Su *et al.* [20] tried to bypassed some of the limitations in Ethereum smart contracts and developed and implemented a blockchain-based supply chain system in the Ethereum platform. Chod *et al.* [21] used the blockchain to more efficiently make inventory transactions and developed a blockchain protocol on Bitcoin to provide supply chain transparency.

We combine the characteristics of blockchain and supply chain finance and select a pledge of warehouse receipts financing and accounts receivable that factor these two scenarios. We use the blockchain transformation of the traditional business processes and solve the previously described current problems in the financial supply chain. In the actual business of financial privacy issues in the supply chain, it is also very important that the proposed scheme will be deployed with a homomorphic encryption algorithm on the blockchain applications. The main contributions of this article are the following two points.

First, we rely on the actual supply chain scenarios of large-scale steel companies and innovatively transform the supply chain finance business model with blockchain technology. We redesign the process and business model of warehouse receipt pledge financing and accounts receivable factoring in supply chain finance. We propose a new supply chain service platform structure based on the blockchain. The platform has been running for one year, has served more than 500 enterprises, and has more than 20 000 certificates. The accumulated transaction amount has reached 1.2 billion dollars. The results show that a business model integrated with a blockchain can effectively solve the existing problems in the traditional supply chain finance process and that the platform effectively improves the efficiency of the business in supply chain finance and reduces the transaction cost. Financing based on the blockchain digital assets not only provides enterprises with convenient and efficient financing channels but also expands the scope of business for warehousing companies.

Second, privacy protection is also a major problem in supply chain finance, where manufacturers' financing quotas, order quantities, transaction prices of intermediate products, etc., are all trade secrets. Companies do not want competitors to know their trading information. The idea for solving the privacy leakage problem in the blockchain technology is closely related to the blockchain architecture and specific application scenarios. Because blockchain relies on public verification, it is contradictory to privacy protection. Furthermore, a universal solution to the problem of privacy disclosure is difficult to find. The usual approach is to find a balance between transaction authentication and privacy protection for a specific scenario. Therefore, combined with the business demand of supply chain finance, we select and improve the Paillier algorithm and embed it into a blockchain network through the smart contract. The test results show that it is a feasible privacy protection scheme on the blockchain.

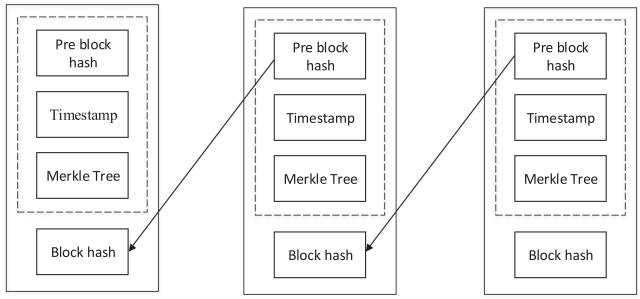


Fig. 1. Block data structure in the blockchain.

In this article, we introduce the blockchain and privacy protection methods in Section II. Then, we introduce the two business models of warehouse receipt pledge financing and accounts receivable factoring and the structure of the platform in Section III. We introduce the details of integrating the homomorphic encryption algorithm into the blockchain in Section IV. Finally, we summarize this article in Section V.

II. Background

A. Blockchain Technology

Blockchain technology was first introduced with Bitcoin. The earliest blockchain definition was a data structure for storing the historical data of Bitcoin transactions [22]. As the scale of the Bitcoin network continues to expand and the market value of Bitcoin continues to grow, the blockchain technology behind Bitcoin has received increasing attention. The Bitcoin system has also become the first representative experiment and application of blockchain technology in the financial field [23]. Fig. 1 shows the block body of Bitcoin. The transaction data are permanently stored in the block structure and sorted by a timestamp. Each block contains information of the previous block's hash; therefore, a chain structure that consists of block data is logically formed. Because of this type of contextual data structure design, the data stored by the blockchain technology are difficult to forge and falsify. The data adopt a distributed storage structure so that the transparency and security of the data are strongly guaranteed. Since the birth of Bitcoin, the scale of the Bitcoin network has continued to expand, and the value of transactions has continued to increase, which validates the effectiveness of the blockchain system design.

Blockchain, as a technology system that comprises a variety of technologies, relies on the traditional distributed databases, cryptography, and network communication technology. The consensus algorithm has a core position and is also a research hotspot. Generally, the efficiency of consensus in distributed systems is low, but the stability of the system is high; the efficiency of consensus in centralized systems is high, but the security and stability of the system are reduced. As a distributed system, blockchain technology needs to consider how to efficiently reach an agreement in a distributed system. Blockchain technology was confined to the public chain architecture in the early stage, and researchers focused on the solution of the Byzantine problem [24]. Satoshi *et al.* proposed that the Bitcoin network uses the

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proof of work (PoW) consensus mechanism to maintain the consistency of the distributed accounting in the Bitcoin network. The core idea is to ensure data consistency and consensus security by introducing computing power competition. With the development of blockchain technology and the emergence of competing currencies, researchers have proposed a variety of mechanisms to achieve network consensus without relying on arithmetic competition, such as proof of stake [25] and delegated proof of stake [26]. In recent years, some consensus mechanisms for the consortium blockchain have been implemented, such as the practical Byzantine fault tolerance and raft consensus algorithms [27]. New consensus algorithms designed for specific application scenarios are still developing rapidly.

Blockchains can be classified as public blockchains or consortium blockchains according to the degree of openness. A public blockchain is a blockchain that is accessible to anyone in an open environment. Anyone can become one of the nodes and make contributions and receive rewards according to the corresponding rules, and no trust relationship exists between the nodes. A public blockchain is completely open and decentralized. At present, a public blockchain is a relatively mature application. In a consortium blockchain, the consensus node is prespecified by the participants. The participants in a consortium blockchain also do not trust one another. Each party chooses its own consensus nodes according to the consensus rules. Transactions in a consortium blockchain must be confirmed by most consensus nodes. The degree of openness and decentralization is weaker in a consortium blockchain than in a public blockchain, but the speed of transaction processing and the waiting time are better in a consortium blockchain than in a public blockchain. Consortium blockchains are used in semiclosed networks, such as the networks built by different enterprises. In such a network, conflicts of interest may exist among different enterprises, and nodes may maliciously tamper with data.

B. Privacy Protection in Blockchain

In the Bitcoin system, transactions are completely exposed. After the transaction data are accumulated, an attacker can conveniently use the big data analysis technology to aggregate and extract a user's transaction address and transaction path, thereby locating the user's identity and learning the user's behavior, which results in a considerable risk of privacy leakage. In this case, it is important to hide the user's trading behavior by obfuscating the transactions. The core idea of the token mixing mechanism is to break up and reorganize multiple transactions to interrupt the direct link between the input addresses and the output addresses in a transaction so that an attacker cannot obtain the user's real transaction data.

Many researchers have focused on privacy protection, and some methods have been used in blockchain. The main methods are the ring signature [28], the blind signature [29], zero-knowledge proof [30]–[32], and homomorphic encryption.

Homomorphic encryption was first proposed in 1978 [33] with the main purpose of creating a set of encryption schemes in which an operation on the encrypted ciphertext is equivalent to perform the same operation on the plaintext before encryption. Therefore, in a scenario, such as cloud computing, only the ciphertext operation is provided to the calculator without revealing the plaintext.

After the concept of homomorphic encryption was proposed, many researchers designed different implementations. Only partial homomorphism is achieved by the original homomorphic encryption schemes, such as the Paillier algorithm that can only perform additive homomorphism [34] and the Elgamal algorithm that can only perform multiplicative homomorphism [35]. The first practical fully homomorphic encryption (FHE) scheme was proposed in 2009. Based on the ideal lattice problem, Gentry divided the FHE scheme into three steps. First, FHE encryption schemes are designed to support only a finite number of homomorphic multiplication and addition operations. Then, the polynomial dimension of the key and ciphertext is reduced by the compression method. Finally, bootstrapping is introduced not only to decrypt the polynomial but also to perform the homomorphic calculation on the ciphertext to produce a new ciphertext. After Gentry, researchers proposed a homomorphic encryption scheme based on the approximate greatest common divisor design in the integer domain, and some improvements to the integer homomorphic encryption scheme have emerged [36].

Many explorations have been made of homomorphic encryption applications, such as secure multiparty computation [37], machine learning [38], and cloud computing privacy protection [39]. In the current application research of homomorphic encryption, efficiency is the focus. Many homomorphic encryption schemes, especially FHE schemes, still cannot satisfy the actual computational speed requirements in terms of execution efficiency.

III. Blockchain-Based Supply Chain Innovation

A. Supply Chain Finance

Supply chain finance is a microfinance concept. Compared with the traditional bank credit loans or venture capital, supply chain finance relies on various production links in the supply chain. Comprehensive financial services must be provided to supply chain stakeholders based on the underlying elements of supply chain production. Supply chain finance can use financial instruments to assist supply chain production and create financial value added through supply chain scenarios to develop good cooperative relationship between the supply chain companies and financial institutions.

China's supply chain finance started late, but due to the rapid development of accounts receivable, commercial paper and financing market businesses have developed rapidly. Supply chain finance management usually involves supply chain core companies, logistics companies, supporting small and medium companies, banks, and other institutions. Different companies and institutions play different roles. According to the information asymmetry theory and the principal-agent theory [40], [41], the participants in supply chain finance screen their own favorable information in business cooperation; therefore, the comprehensiveness of the business information cannot be guaranteed. Only by ensuring the full exchange and credibility of the financial cooperation information in the supply chain, we can

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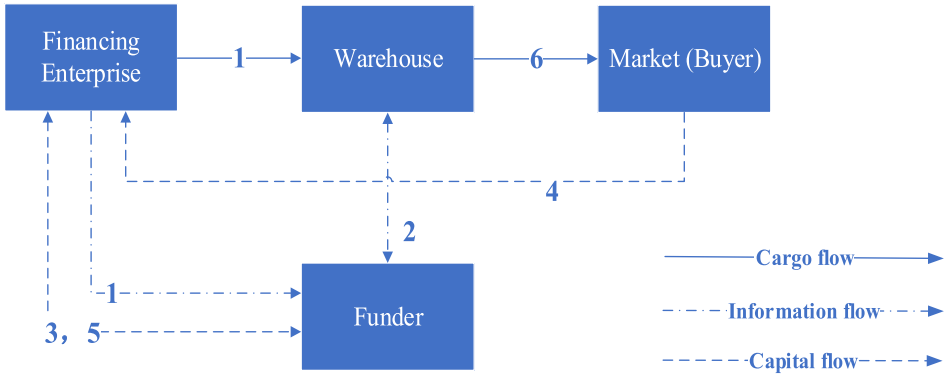


Fig. 2. Traditional warehouse receipt pledge financing business model.

ensure the efficiency of capital flow and logistics. Some common problems in current supply chain finance are concentrated on the credit risk of supply chain finance. In the current supply chain financial system, trusted data are difficult to obtain, and risks are difficult to control. Therefore, a platform must be built to achieve data sharing.

Blockchain technology can be used to build a supply chain financial platform to effectively solve the problem of insufficient data sharing. The technical characteristics of the blockchain bring convenience to auditing and supervision and are also beneficial to the risk control of the supply chain finance business. We use a blockchain to build a supply chain financing platform. First, we select two blockchain-based scenarios, specifically, warehouse receipt financing and accounts receivable financing, as examples to introduce the change of the business model, process, and benefits after using the blockchain in detail. Then, we introduce the technical framework and each module's function of the blockchain-based supply chain system and compare the platform with other platforms.

B. Warehouse Receipt Financing

The main purpose of the traditional warehouse receipt financing is to solve the short-term capital demand in the production process of an enterprise. Participants in this model include financing companies, warehouses, sales markets, and funders. The specific cargo flow, capital flow, and information flow are shown in Fig. 2.

1) The financing company issues a financing request to the funder and submits the financing target to the warehouse in accordance with the agreement.
2) The warehouse audits the financing target and submits the warehouse receipt to the funder. The warehouse will then manage the target in strict accordance with the requirements of the funder.
3) The funder provides financial support to the financing company after reviewing the warehouse receipt.
4) The financing company sells the goods and obtains sales proceeds from the market.
5) The financing company pays the principal and interest of the funder.

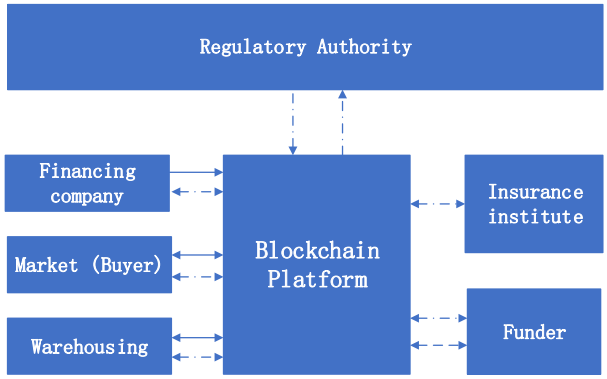


Fig. 3. Warehouse receipt pledge financing business model based on the blockchain.

6) The warehouse pays the product to the buyer of the sales market under the authority of the funder.

However, the most important problem of the existing electronic warehouse receipt transactions platform is that the current electronic trading system cannot guarantee the authenticity of the transaction data and the validity of the electronic warehouse receipts of the transaction.

The transaction system is complicated, and the transaction personnel is difficult to coordinate. Therefore, we use blockchain in the supply chain financial service platform as the data bus to redesign the business structure and replan the information flow, cargo flow, and capital flow of the warehouse receipt pledge financing business, as shown in Fig. 3. The entire system can connect the warehouse receipt pledge financing business parties, including financing applicants, warehouses, regulatory authorities, the blockchain-based warehouse receipt management platform, digital asset centers, and fund providers. The process of warehouse receipt pledge financing is shown in Fig. 4.

1) The customer uses the existing system to initiate a financing request, generates a financing contract draft, and saves it to the current system; the system simultaneously initiates a blockchain transaction, which records key information (customer ID, transaction ID, etc.) and the fingerprint of the financing contract draft. At the same time, blockchain triggers the warehouse contract to audit.

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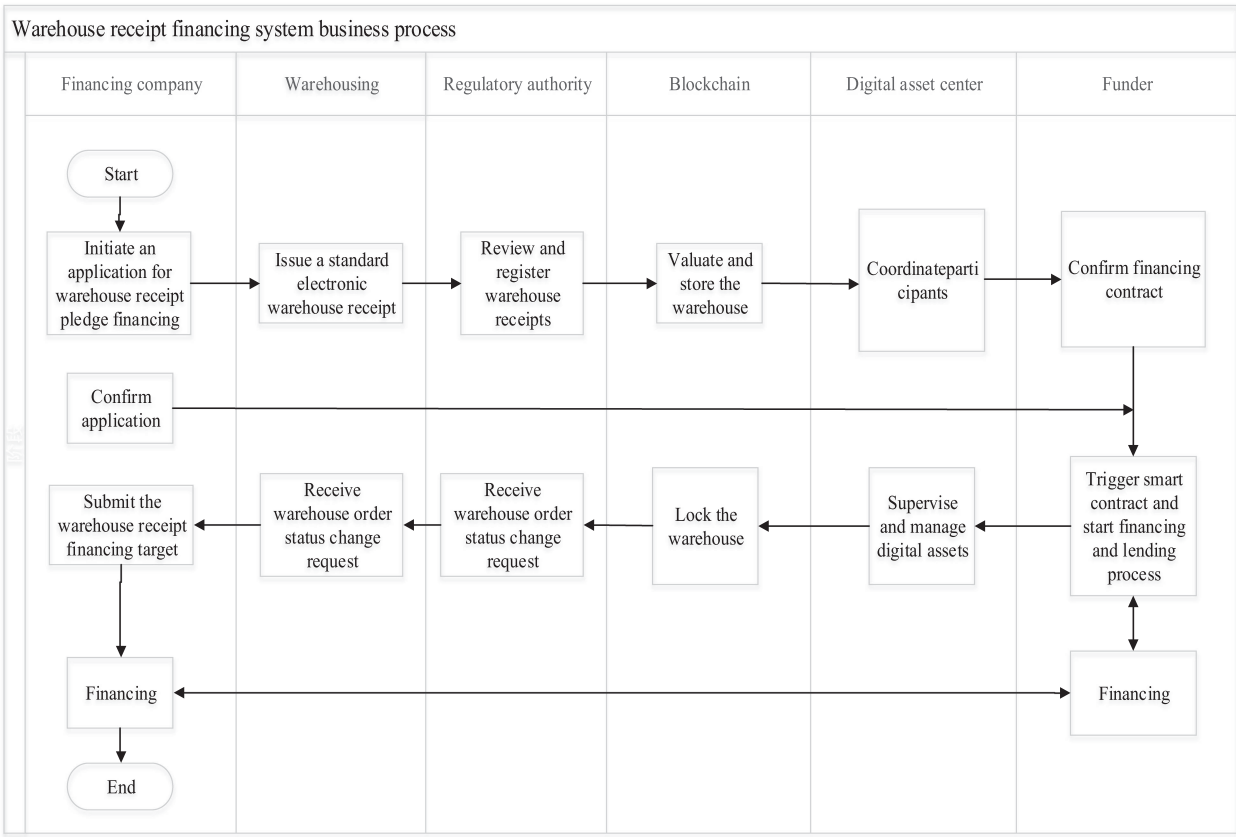


Fig. 4. Warehouse order pledge financing business process based on the blockchain.

2) The warehouse contract is triggered by the financing transaction application. The contract follows the auditing steps, compares the customer input information with the information on the blockchain, draws a recommended conclusion (approved, failed), and writes the result into the blockchain.
3) The regulatory authority reviews and registers the conclusions drawn from the warehouse contract and records the results on the blockchain.
4) The warehouse contract checks the audit results and gives the warehouse order valuation.
5) The insurance staff queries the existing business system and observes the customer's financing request, warehouse review progress, results, etc., as well as the blockchain data information (including the current financing information, warehouse audit results, historical financing situations, and results of the audit).
6) The funder confirms the financing contract according to the dispatch of the digital asset management center and triggers the smart contract to start the lending process. The issuance and recovery of financing funds are conducted directly between the funder and the financing applicant. The status change of the corresponding warehouse receipt assets is updated among the departments to realize the entire process management of the warehouse receipts and to avoid the occurrence of repeat warehouse receipts and other issues.

C. Accounts Receivable Factoring

Any production enterprise will produce accounts receivable in production and operation activities, and the management of accounts receivable is related to the liquidity and efficiency of the enterprise. The essence of an enterprise's accounts receivable is that the liquidity of the enterprise is occupied by its customers, but the enterprise itself has to bear the cost of occupying this part of the liquidity.

Accounts receivable factoring is a way of using receivable financing and is a very active account receivable management model in the current financial market. In the accounts receivable factoring business, the financing enterprise will transfer the accounts receivable formed by credit sales to the factoring agent according to certain conditions and obtain the corresponding financing amount.

Since accounts receivable factoring development has a relatively short history in China and there is also a lack of a sound credit trading system in China, accounts receivable factoring in China has a relatively large risk.

This also results in the scope of the accounts receivable factoring product range being relatively small. As shown in Fig. 5, three main roles exist in the accounts receivable factoring business model, namely, the financing company (seller), the buyer, and the factoring company. China's factoring is usually a bank, and the factoring business is similar to a bank's loan business. The factoring business plays an important role

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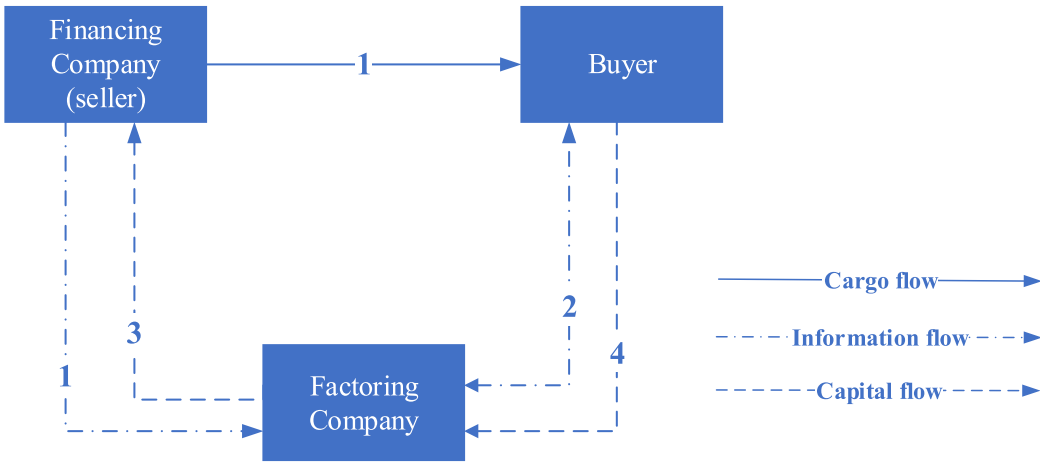


Fig. 5. Accounts receivable factoring model based on the blockchain.

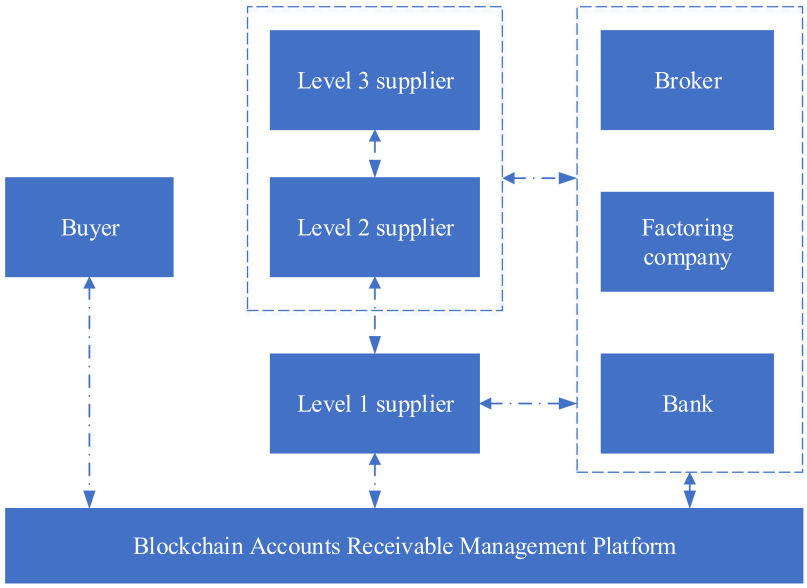


Fig. 6. Accounts receivable factoring model based on the blockchain.

in the management of accounts receivable, especially in the management of the accounts receivable of the small and medium companies.

To this end, we have used blockchain technology to build a reliable platform to manage accounts receivable factoring. The business structure is shown in Fig. 6.

In the blockchain-based factoring model, the transfer of receivable bonds requires notification to the core company and the signing of a confirmation, which is decentralized and requires multiple confirmations. Automatic cycle management through intelligent contracts based on the blockchain technology can reduce the links in the accounts receivable factoring mode and improve system efficiency. Fig. 7 shows the blockchain-based business process. The accounts receivable factoring business relies on the credit sales of goods. The supplier sells goods in the form of credit sales. After the buyer agrees to the purchase and gives the payment commitment, the financing application can be submitted. After the supervisory authority reviews the buyer and seller qualifications in the sale of the goods, the accounts receivable is digitally astatized and submitted to the factoring company. The factoring business begins the factoring process based on the corresponding accounts receivable. Once the factoring business begins, the main body of credit and debt management is transformed from the suppliers and buyers to the buyers and factoring companies. During the factoring business, the factoring company needs to pay the supplier the financing amount of the factoring financing, and at the same time, it is necessary to urge the buyer to pay the money after the sale of the goods and bear the corresponding risk of capital recovery. The regulator and digital asset management center only need to track the status of the factoring business.

The accounts receivable factoring business model with the introduction of blockchain is described as follows.

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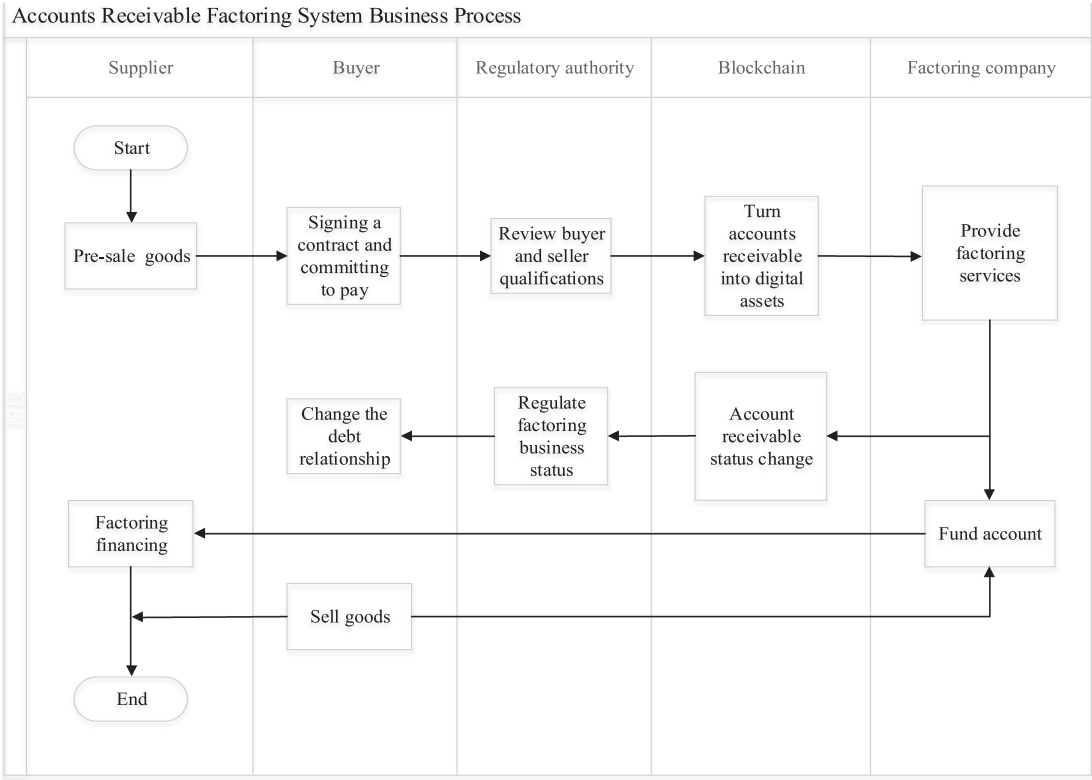


Fig. 7. Accounts receivable factoring business process based on the blockchain.

1) As a fund provider, the factoring company completes the credit evaluation of the suppliers and core enterprises through the blockchain credit management module.
2) The core enterprise and the supplier form a smart contract for the purchase and sale contract and agree on the information, such as the payment account and the accounting period. The supplier initiates a factoring application to the factoring company in the system. The smart contract automatically transfers the account bond to the lending institution, modifies the payment account, and informs the core enterprise. If the accounts receivable is overdue, the core company automatically pays the account. Then, a transfer is made to the account of the lending institution.
3) After the supplier initiates the factoring application, the smart contract confirms the information and records the factoring contract information and the accounts receivable transfer list. Then, the smart contract issues the lending funds from the lending institution's account to the designated account of the factoring contract.
4) The core enterprise automatically pays the accounts receivable when it expires, and the smart contract settles the principal and interest according to the set and transfers the remaining funds to the designated account of the supplier.

D. Structure of the Platform and the Analysis

In the supply chain finance, each enterprise and user involved has different credibility. Core enterprises, banks, and regulators have higher credibility than other small enterprises and ordinary users. Different from the public blockchain, in consortium blockchain, we need users with higher credit as verification nodes to be responsible for the transaction confirmation and consensus. Therefore, we choose some core organizations, such as core enterprises, banks, and regulators, to maintain the blockchain nodes. The other small- and medium-sized enterprises can access the blockchain-based supply chain finance platform through the interface or directly connect the corresponding core enterprises. Fig. 8 shows the development structure of the blockchain-based platform.

The application of blockchain technology in the traditional field is often an innovation based on the traditional system. Taking our platform as an example, blockchain mainly plays an important role in the underlying data storage and management module. Some modules in the traditional system are still indispensable. These modules can be directly integrated on the blockchain in the form of smart contracts or can be connected with the blockchain platform through the interface. Fig. 9 shows the structure of the supply chain finance platform. The platform logically adopts a six-layer structure and different logical layers correspond to the different system function modules.

1) The presentation layer is used to directly interact with the system user and can be shown as a web page or a mobile application.
2) The auditing supervision layer constructs a security barrier between the display layer and the business logic layer and can shield the malicious operation of the user from

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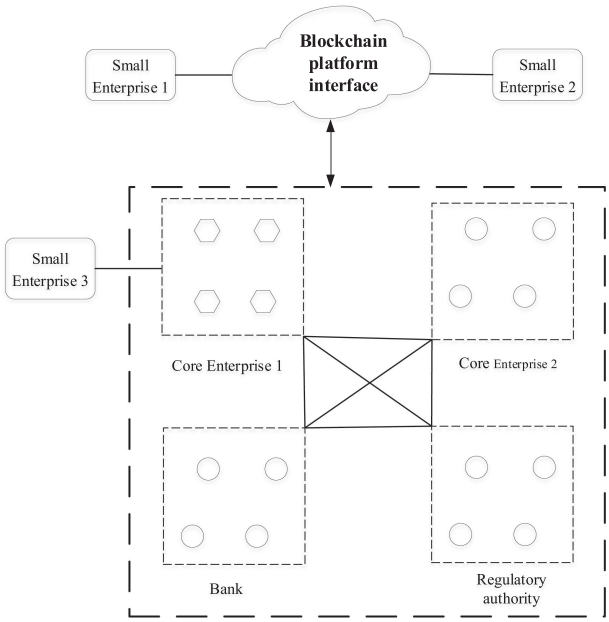


Fig. 8. Development structure of the platform.

being unsafe and can internally check the compliance of the business logic.

3) The business logic layer performs the submodule management after integrating the services in supply chain finance.
4) The public service layer abstracts the characteristics common to the business logic layer and implements unified implementation, thereby increasing the reusability of the system modules.
5) The data management layer provides database services for the entire platform and is responsible for the management of all underlying data.
6) The underlying hardware layer is the hardware support for the entire system, including servers for information systems and IoT sensors for collecting data.

Our platform mainly solves the financial problems in the supply chain. There are few existing pieces of literature detailed scheme introduction in this field. Therefore, we select several blockchain platforms in the supply chain field for comparison and Table I presents the comparison result. Most papers used Bitcoin or other public blockchains as the underlying blockchain in their supply chain innovation. However, it is not a good choice to use a public chain in a supply chain because of poor performance and high cost of a public blockchain. We use a consortium blockchain as the underlying chain structure. The transactions per second are more than 1000, and the delay of confirming transactions is approximately 4 s. A consortium blockchain is much better than a public blockchain. In addition, it is easier to design the privacy protection module on a consortium blockchain.

Under the traditional supply chain financial model, a core enterprise is usually required to coordinate as a central node. The traditional financing business is difficult to operate normally in the absence of a core enterprise. Through the multicenter nature of the blockchain system, a corporate alliance can be built to accomplish cooperation. The number of institutions involved in supply chain finance has increased rapidly. For example, financial institutions are no longer limited to factoring companies. Financial institutions, such as banks and brokerage firms, can participate in additional financial operations based on the accounts receivable or accounts receivable factoring.

With the blockchain technology, the supply chain finance service platform can provide a reference for the construction of the credit system of the supply chain through the smooth supply chain information channel. The technical characteristics of the blockchain ensure the authenticity and validity of the information flow, which facilitates the transfer of credit.

The data are more reliable. The data stored in the blockchain structure cannot be tampered with once formed, thereby increasing the difficulty of fraud by a single party. The entire business process is open and transparent, and each interested party checks, balances, and supervises one another. The introduction of government regulatory authorities and insurance institutions can effectively prevent collusion among institutions and reduce the risks and regulatory costs.

The information flow and capital flow are integrated. The issuance and recovery of funds can be performed in the blockchain system through the establishment of the blockchain warehouse receipt public account. This combination of the capital flow and information flow can substantially reduce the complexity of fund account collection, increase the speed of the capital flow, and provide effective regulatory audit conditions for regulators.

IV. Blockchain Privacy Protection in Supply Chain Financial Scenarios

Blockchain was originally derived from Bitcoin, so many of the technical features in blockchain technology are directly or indirectly affected by Bitcoin. In a centralized trading system, the strategy for privacy protection is to determine the identity but hide the transaction. It is difficult for nodes other than the centralized node to fully know all transactions of a specific account. The original blockchain trading system represented by Bitcoin is essentially an open and distributed accounting system. All accounts and trading information are transparent, and different privacy models are, thus, needed. The strategy of these blockchain trading systems is to hide the identity but determine the transaction by using a decentralized system in which electronic identities do not have to correspond to real identities. A real-world individual can have multiple account addresses without his real identity to hide his operations.

Although the Bitcoin-like system protects the real identity of the transaction address owner, different account addresses can be correlated by analyzing the publicly traded data in a certain way, thereby infringing on a user's privacy [42]. After the emergence of the consortium chain architecture, the submission of transactions requires the identification of the consortium member's signature, thereby invalidating the privacy protection model that relies on the true identity of the owner of the protection transaction address. This process will cause a privacy leak because a member of the alliance must submit a transaction with sensitive data. To make the consortium chain more available, the privacy

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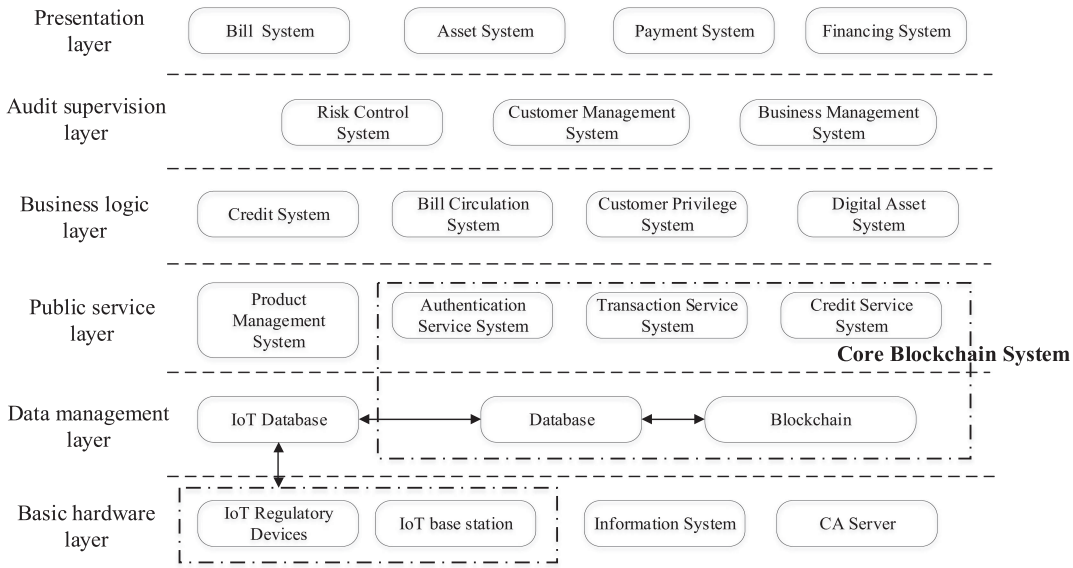


Fig. 9. Supply chain financial model based on the blockchain.

TABLE I
COMPARISON OF CURRENT PLATFORMS

	POMS [18]	*b verify [21]*	*SmartSupply [20]*	*Our platform*
Blockchain choice	Bitcoin	Bitcoin	Ethereum	Consortium blockchain
Tamper-resistant	Yes	Yes	Yes	Yes
Transaction privacy protection	No	No	No	Yes
Consensus	PoW	PoW	PoW/BFT	BFT
Pressure of the main chain	Greater	Greater	Less	Less
Difficulty of system implementation	Difficult	Easier	Easier	Easiest
System cost	High	High	Low	Low
Delay of transactions confirming on the blockchain	1h	1h	15s	5s
Transactions per second	10+	10+	100+	1000+

protection model should be studied from a new perspective, and a privacy protection algorithm must be designed for sensitive data operations in the consortium chain.

The core contradiction of privacy protection in a consortium blockchain is that the chain still relies on the publicity of information to obtain the endorsement of other members of the consortium, which results in information leakage. The ciphertext is the only information that the traditional encryption method can reveal, and other people cannot crack the ciphertext. However, the encrypted ciphertext information cannot be normally changed; otherwise, the decryption will fail. Therefore, the encrypting of information makes it difficult to process the information, which limits the application of the consortium chain. We want to introduce an encryption method in the consortium chain. Such a method should make it difficult to crack the ciphertext while retaining the ability to process the encrypted information.

A homomorphic encryption scheme should be designed according to the actual needs. The current FHE algorithm still has practical efficiency problems; therefore, a partial homomorphic encryption algorithm is more feasible. The number of transactions and the number of products in the supply chain finance business are usually sensitive. Accordingly, this article modifies the Paillier algorithm to integrate it into the blockchain system and tests the algorithm in a laboratory environment.

A. Paillier Algorithm in Blockchain

The Paillier algorithm is based on the high-order residual class problem and has additive homomorphism. Many useful privacy protection algorithms can be constructed based on the Paillier algorithm.

Definition 1: Assuming that p, q are two large prime numbers and that $n = p * q$, there are a Euler function and a Carmichael function (lcm represents the least common multiple)

$$\begin{aligned} \varphi(n) &= (p-1)(q-1) \\ \lambda(n) &= \operatorname{lcm}(p-1, q-1) \\ |Z_{n^2}^*| &= \varphi\left(n^2\right) = n\varphi(n) \quad \forall \omega \in Z_{n^2}^*. \end{aligned}$$

Thus

$$\omega^{\lambda} \equiv l \bmod n, \quad \omega^{n\lambda} \equiv l \bmod n^2.$$

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Definition 2: Assume that p, q are two large prime numbers, $n = p * q$ and $g \in Z_{n^2}^*$.

Define the function ε_ϵ as

$$Z_n \times Z_n^* \to Z_{n^2}^*$$
$$(x, y) \to g^x \cdot y^n \bmod n^2.$$

When the order of g in group $Z_{n^2}^*$ is a multiple of n, and ε_ϵ is a bijection mapping. For a given $\omega \in Z_{n^2}^*$, $x \in Z_n$, we have

$$\exists y \in Z_n^*$$
$$\varepsilon_g(x, y) = \omega.$$

Given ω, n, and g, it is difficult to calculate $\varepsilon_g(x, y)$, but it is easy to calculate $\varepsilon_g(x, y)$, given λ. Thus, we can design an encryption approach based on this method.

The Paillier encryption algorithm is divided into three stages of generating the encryption key, encryption, and decryption.

1) Generating the encryption key.
   a) choose two large prime numbers p and q such that the following equation holds (gcd is the greatest common divisor):

   $$\gcd(pq, (p-1)(q-1)) = 1.$$

   The equation holds when the two prime numbers are equal in length.
   b) calculate λ:

   $$n = pq$$
   $$\lambda = \operatorname{lcm}(p-1, q-1).$$

   c) choose $g \in Z_{n^2}^*$ randomly:
   d) define

   $$\mathrm{L}(x) = \frac{x-1}{n}$$
   $$\forall \mu \in \{\mu \langle n^2 \mid \mu \equiv l \bmod n\}$$

   subject to

   $$\mu \equiv \left(L\left(g^\lambda \bmod n^2\right)\right)^{-1} \bmod n.$$

   Thus, we can obtain the public key

   $$(n, g)$$

   and private key

   $$(\lambda, \mu).$$

2) Encryption.
   The plaintext must satisfy

   $$0 \le m < n.$$

   Choose a random number *r* subject to

   $$0 < r < n$$
   $$r \in Z_{n^2}^*.$$

   Calculate the ciphertext

   $$c = g^m \cdot r^n \bmod n^2.$$

3) Decryption.
   The ciphertext should satisfy

   $$\mathrm{c} \in Z_{n^2}^*.$$

   Calculate the plaintext

   $$m = L\left(c^\lambda \bmod n^2\right) \cdot \mu \bmod n.$$

   A Paillier algorithm designed in this way can achieve the following homomorphic operation characteristics.

1) Adding two ciphertexts.

   $$\mathrm{D}(\mathrm{E}(m_1, r_1) \cdot \mathrm{E}(m_2, r_2) \bmod n^2) \equiv m_1 + m_2 \bmod n.$$

2) Adding a ciphertext and a plaintext.

   $$\mathrm{D}\left(\mathrm{E}(m_1, r_1) \cdot g^{m_2} \bmod n^2\right) \equiv m_1 + m_2 \bmod n.$$

3) Multiplication of ciphertext and plaintext.

   $$\mathrm{D}\left(\mathrm{E}(m_1, r_1)^{m_2} \bmod n^2\right) \equiv m_1 m_2 \bmod n.$$

The Paillier algorithm can usually be applied only to non-negative integers, but in the blockchain supply chain financial scenario, an algorithm that can handle only non-negative integer operations is difficult to use. To integrate the Paillier homomorphic encryption algorithm into the blockchain supply chain financial scenario, the algorithm must be optimized to handle signed integers and signed float numbers.

Consider a public key (*n, g*) and a private key (λ, μ) generated by a Paillier algorithm. We can use the mathematical characteristics of the algorithm to encode before encryption and decode after decryption to implement signed integers and signed float numbers.

1) Encode:
   a) if a number m' to be encoded is a float number, multiply it by b^{-x}, $b \in 2^y, y \in Z^+$, then round it to represent the value as an integer m:
   b) the m must satisfy $\operatorname{abs}(m) < n/3$. If $m < 0$, $m = m + n$.
2) Decode:
   a) for a decoded number e, the original number should be positive if $e < n/3$ and negative if $e > 2n/3$. We subtract n from e, if $e > 2n/3$:
   b) decode the origin number with the equation $e \times b^x$.

Usually, the base b and exponent x are judged according to the actual situation. Multiplication by a larger number can produce higher coding precision but compresses the range of values of the coded float number. Multiplication by a smaller number can produce a larger range of float numbers. Fig. 10 shows the improved algorithm flow.

B. Combination of the Homomorphic Encryption Algorithm and Blockchain

We have chosen the open-source consortium blockchain, Hyperledger fabric, as the test platform. The homomorphic encryption algorithm relies on smart contracts to embed in

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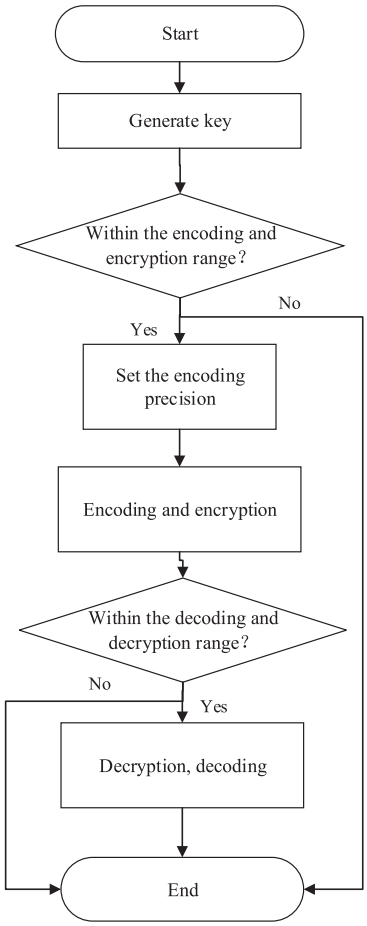


Fig. 10. Flowchart of the improved Paillier algorithm.

a blockchain network. Hyperledger fabric's smart contract, namely, chaincode, is responsible for interactive functions, such as the verification, storage, and execution of the chain operations. These smart contracts are divided into two categories as follows: the system smart contract, which runs on the peer node, is not subject to the life cycle and is used mainly to implement the system-level behavior; and the normal contract, which runs in a separate container, is subject to the life cycle and provides the calling service in the form of an interface.

A Paillier system chaincode (PSCC) can be designed and integrated into the consortium blockchain.

The main components of PSCC are the following.

1) Key generation: Generate a pair of public and private keys for encryption.
2) Encryption: Encrypt the plaintext.
3) Decryption: Decrypt the ciphertext.
4) Encode: Encode a signed integer or float number.
5) Decode: Decode an encoded number.

When the service system that relies on the consortium blockchain needs to process a service involving sensitive data, PSCC can be called to complete the corresponding service processing, as shown in Fig. 11.

1) The application requests an endorsement from the endorser node for this transaction.
2) The endorser node compares the permissions of the application and signs and generates a signed READ–WRITE set.
3) The endorser node replies to the application according to the protocol and returns the generated READ–WRITE set.
4) The application is ready to submit the transaction data. Then, PSCC performs the homomorphic encryption of sensitive data, and notarized publicity data are requested.
5) A homomorphic operation is performed on the two pieces of data in 4), and a transaction data stream is generated.
6) The ordering node orders the transaction and packs it into blocks.
7) The committer node obtains the packed data block from the ordering node and performs legality verification on the data block.
8) PSCC decrypts the data and then submits the transaction data to the ledger.

Through this process, the contradiction between the authentication and privacy protection of the transaction data in the consortium chain can be resolved. That is, the transaction data are split, and the part of the authentication that needs to be publicized is still publicized, while the private data can be hidden by homomorphic encryption. This design is related to the technical characteristics of the blockchain system. Because the blockchain system originated from Bitcoin, the account model in the Bitcoin design is based on the unspent transaction output data structure. In essence, the blockchain system is notarized not only for the variables in the transaction process but also for the entire transaction process. PSCC can guarantee that private information is not revealed while notarizing the transaction process.

C. Performance of PSCC

The length of the key pair used during PSCC has a substantial impact on operational efficiency. We evaluated PSCC's performance on a server with an i7-4790 CPU, 8 GB RAM and 500 GB HDD. Table II gives the average time required for a key pair generation. As the length of the key pair increases, the time required to generate the key pair increases considerably. The generation time of a key pair with a length of fewer than 512 bits is on the order of milliseconds, and the generation time of a key pair with a length greater than 512 bits is on the order of seconds. From a production perspective, considering the efficiency and security balance, a key pair with a length of 256 or 512 bits is ideal.

Table III gives the time required for the encryption and decryption of different key pairs of different lengths. As the length of a key pair increases, the time required for encryption and decryption increases rapidly. For a given key length, the time required for encryption is usually greater than the time required for decryption. From a practical perspective, keys with a length of 256 or 512 bits are suitable for the supply chain scenario. Table IV gives the relationship between the key pair length and the time required for plaintext and ciphertext calculations. As the key length increases, the computation time increases rapidly. In general, for a given key length, the addition operation is faster than the multiplication operation.

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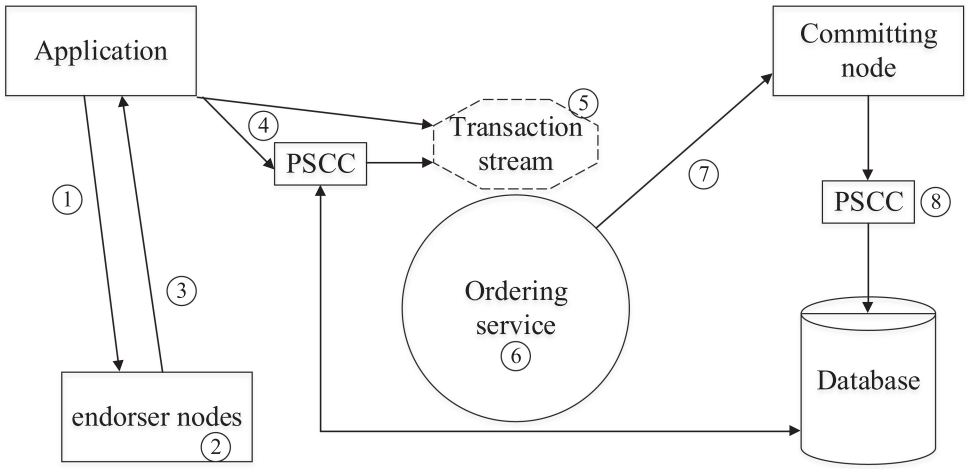


Fig. 11. Transaction life cycle process with PSCC.

TABLE II
TIME REQUIRED FOR KEY PAIR GENERATION

Number of key pairs (bits)	*Average time (s)*	*Number of generations per second*
128	0.003641	274
256	0.009805	101
512	0.031278	31
1024	0.143347	6
2048	1.004862	1
4096	9.003019	0.11
8192	51.519110	0.19

TABLE III
TIME REQUIRED FOR ENCRYPTION AND DECRYPTION

Number of key pairs (bits)	*encryption*	*decryption*
128	0.000082	0.000052
256	0.000374	0.000177
512	0.001931	0.000728
1024	0.012756	0.004035
2048	0.086810	0.025335
4096	0.606871	0.174780
8192	4.421056	1.233789

In the supply chain finance business, more plaintext–ciphertext operations and ciphertext operations are used in homomorphic encryption, and multiplication operations are less common. As given in Table IV, the times required for plaintext plus ciphertext and ciphertext plus ciphertext operations are not substantially different, and both are much smaller than the times required for the encryption and decryption.

As shown in Fig. 12, a key pair with a length of 512 bits is taken as an example. Assume that a PSCC process has the following four steps: generating a key pair, encrypting, performing ciphertext plus plaintext addition operations, and decrypting operations. The key pair generation time accounts for the largest proportion of the entire process, followed by the encryption time and the decryption time, and the addition operation time is the smallest.

According to the experimental results, we can draw the following conclusions.

1) PSCC should select the key pair length according to the throughput of the consortium blockchain. A key pair length of 256 or 512 bits is appropriate.
2) Because the generation of key pairs takes a long time, a key pair reuse mechanism should be designed. We can reduce the entire process time by designing a unified key management system.
3) Homomorphic encryption is inefficient and is not suitable for large data operations. However, in the blockchain supply chain financial scenario, considering the speed of the blockchain and the specific business process, PSCC can still satisfy the scenario application requirements. When designing a business process, it is necessary to minimize the number of homomorphic encryption operations to avoid the computational speed of homomorphic encryption becoming a system bottleneck.

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TABLE IV
TIME REQUIRED FOR HOMOMORPHIC OPERATIONS

Number of key pairs (bits)	*Plaintext+ Ciphertext (s)*	*Ciphertext + Ciphertext (s)*	*Ciphertext*plaintext (s)*
128	0.000006	0.000004	0.000037
256	0.000007	0.000005	0.000095
512	0.000008	0.000008	0.000253
1024	0.000014	0.000018	0.000849
2048	0.000030	0.000060	0.002868
4096	0.000180	0.000240	0.010234
8192	0.000501	0.000600	0.037179

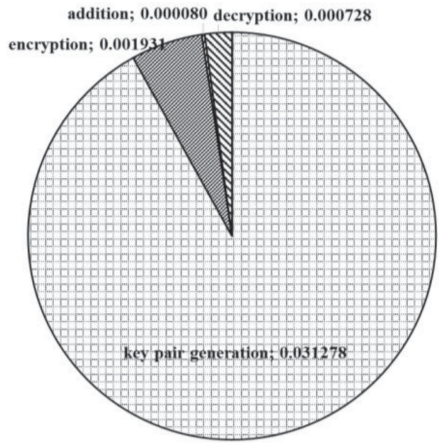


Fig. 12. Time required for each part of a PSCC process.

V. CONCLUSION

Supply chain finance, an important part of the field of economics, faced difficult problems of trust and information asymmetry. These problems had also seriously hampered the enthusiasm for supply chain finance development. In this article, we built a blockchain-based supply chain financial service platform to solve the problems of the traditional supply chain finance. We selected two blockchain-based scenarios, namely, the warehouse receipt financing and accounts receivable financing, as examples to introduce in detail the change of the business model and process after using the blockchain. It tells the organizations who want to use the blockchain in supply chain financing that how to design and use the blockchain in this area. Then, we focused on the privacy protection and proposed an experimental feasible homomorphic encryption scheme on the blockchain suitable for the supply chain financing scenario.

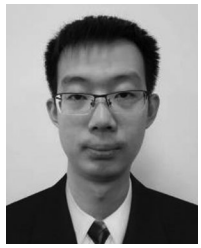
The platform had been running stably for approximately one year in actual scenarios and had served more than 500 supply chain companies. The accumulated transaction amount has reached 1.2 billion dollars. The platform established a peer-to-peer trust mechanism, which enables companies in the industry chain to participate more equitably in the business. This process builds an environment of trust and integrity, which was conducive to achieve a win–win scenario for each company in the supply chain. In the future, we will continue to research homomorphic encryption and the blockchain structure. We will attempt to apply the privacy protection scheme mentioned in this article to the actual production process and realize high-efficiency full homomorphism in the blockchain.

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